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Mask

Abstract

A mask may be configured as a respirator. A mask may be relatively easy to manufacture. A mask may be reusable and customizable. A mask may include a strap attachment configured such that a strap is attachable to the strap attachment without tying the strap into a knot.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to a mask, and in particular a mask configured as a respirator.

BACKGROUND

(2) Respirators are a type of mask worn over the mouth and nose to protect the wearer's respiratory system by filtering out substances such as dust, fumes, bacteria, etc., from inhaled air.

SUMMARY

(3) In some aspects, the techniques described herein relate to a mask, including: a strap attachment configured such that a strap is attachable to the strap attachment without tying the strap into a knot.

(4) In some aspects, the techniques described herein relate to a mask, wherein the strap attachment includes a first through-hole, a second through-hole, and a third through-hole configured such that the strap is attachable to the strap attachment by passing the strap through each of the first, second, and third through-holes.

(5) In some aspects, the techniques described herein relate to a mask, wherein: the first through-hole is spaced-apart from the second through-hole in a first direction, and the third through-hole is spaced-apart from the first and second through-holes in a second direction perpendicular to the first direction.

(6) In some aspects, the techniques described herein relate to a mask, wherein the strap attachment is configured such that the strap is attachable to the strap attachment by passing a free end of the strap out of the third through-hole and underneath a segment of the strap between the first and second through-holes.

(7) In some aspects, the techniques described herein relate to a mask, wherein the strap attachment includes an indicator arrow between the first and second through-holes pointing in a direction away from the third through-hole.

(8) In some aspects, the techniques described herein relate to a mask, wherein the first, second, and third through-holes are elliptical and exhibit major axes that are non-parallel to one another.

(9) In some aspects, the techniques described herein relate to a mask, wherein the strap is a strand of elastic material having first and second free ends.

(10) In some aspects, the techniques described herein relate to a mask, wherein: the strap attachment is a first strap attachment and the mask further includes second, third, and fourth strap attachments, the first and second strap attachments are configured to attach to a first strand without tying the first strand into any knots, and the third and fourth strap attachments are configured to attach to a second strand without tying the second strand into any knots.

(11) In some aspects, the techniques described herein relate to a mask, wherein the strap attachment includes a through-hole configured to receive a fastener.

- (12) In some aspects, the techniques described herein relate to a mask, wherein: the fastener includes a head, a shaft, and a deflectable loop, and the deflectable loop is biased to a relaxed state in which the deflectable loop is able to retain the strap attachment on the shaft.
- (13) In some aspects, the techniques described herein relate to a mask, wherein: the head is arranged between an inner shell and an outer shell of the mask, and the shaft projects through an opening in the outer shell.
- (14) In some aspects, the techniques described herein relate to a mask, further including: a filter arranged between the inner and outer shells.
- (15) In some aspects, the techniques described herein relate to a mask, wherein: the outer shell includes a first opening in a first side wall, a second opening in a second side wall opposite the first side wall, and a third opening in a bottom wall, the inner shell includes a first opening in a first side wall, a second opening in a second side wall opposite the first side wall, and a third opening in a bottom wall such that the first, second and third openings of the inner shell substantially align with corresponding first, second and third openings of the outer shell, and the filter overlaps the first, second, and third openings of the inner and outer shells.
- (16) In some aspects, the techniques described herein relate to a mask, wherein the inner and outer shells each include: a forward panel; and a perimeter wall projecting from an edge of the forward panel to a free end.
- (17) In some aspects, the techniques described herein relate to a mask, wherein the perimeter walls of the inner and outer shells are tapered toward the forward panels.
- (18) In some aspects, the techniques described herein relate to a mask, wherein: the first, second, and third openings of the outer shell are formed in the perimeter wall of the outer shell, and the first, second, and third openings of the inner shell are formed in the perimeter wall of the inner shell.
- (19) In some aspects, the techniques described herein relate to a mask, further including: a seal adjacent the free ends of the perimeter walls of the inner and outer shells.
- (20) In some aspects, the techniques described herein relate to a mask, wherein the forward panels are transparent.
- (21) In some aspects, the techniques described herein relate to a mask, wherein the filter is arranged so as to not obscure a view through the forward panels.
- (22) In some aspects, the techniques described herein relate to a mask, including: an outer shell including a first opening in a first side wall, a second opening in a second side wall opposite the first side wall, and a third opening in a bottom wall; an inner shell including a first opening in a first side wall, a second opening in a second side wall opposite the first side wall, and a third opening in a bottom wall such that the first, second and third openings of the inner shell substantially align with corresponding first, second and third openings of the outer shell; and a filter overlapping the first, second, and third openings of the inner and outer shells, wherein the inner and outer shells each include a transparent forward panel and the filter is arranged so as to not obscure a view through the forward panels.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front perspective view of an example mask worn by a wearer.
- (2) FIG. 2 is an exploded view of the example mask.
- (3) FIG. 3 is a front view of an inner shell of the example mask.
- (4) FIG. 4 is a bottom view of the inner shell.
- (5) FIG. 5 is a rear-perspective view of the inner shell.
- (6) FIG. 6 is a front view of an outer shell of the example mask.

(7) FIG. 7 is a bottom view of the outer shell.

(8) FIG. 8 is a side view of an example fastener.

(9) FIG. 9 is another side view of the example fastener.

(10) FIG. 10 is a top view of an example strap attachment.

(11) FIG. 11 is a perspective view of the strap attachment arranged relative to the example fastener and a strap.

DETAILED DESCRIPTION

(12) This disclosure relates to a mask, and in particular a mask configured as a respirator. Among other benefits, which will be appreciated from the below description, the disclosed mask is relatively easy to manufacture. Further, the disclosed mask is reusable, customizable, and adjustable.

(13) FIG. 1 illustrates a mask 10 being worn by a wearer 12. The mask 10 includes a number of components, and thus the mask 10 may be referred to as a mask assembly, mask system, a system, or an assembly. The mask 10 is configured as a respirator, and, when worn by the wearer 12, is configured to seal against a face of the wearer 12 so as to cover the mouth and nose of the wearer 12. In doing so, the mask 10 protects the respiratory system of the wearer 12 by filtering out substances such as dust, fumes, bacteria, etc., from inhaled air. The mask 10 also filters air exhaled by the wearer 12 to protect others in close proximity to the wearer 12 from any substances that may be present in the air exhaled by the user.

(14) With joint reference to FIGS. 1 and 2, the mask 10 includes a facepiece portion 14 configured to surround a nose and mouth of the wearer 12. Adjacent the face of the wearer 12, the facepiece portion 14 is attached to a seal 16 which is configured to directly contact and seal against the face of the wearer 12. First and second straps 18A, 18B attached to the facepiece portion 14 pass around a head and/or neck of the wearer 12 and hold the seal 16 against the face of the wearer 12. In an example, the straps 18A, 18B are each provided by a respective strand of elastic material having two free ends and without any closed loops. In a particular example, the straps 18A, 18B exhibit a hollow, non-circular cross-section, which facilitates attachment to the mask 10 without tying knots.

(15) With reference to FIG. 1, the facepiece portion 14 includes a front viewport 20, which is essentially a transparent window in a forward-facing portion of the facepiece portion 14 through which a view of the mouth and nose of the wearer 12 is unobstructed. The mask 10 is configured such that the front viewport 20 is spaced-apart from the tip of the nose of the wearer 12 and the mouth of the wearer 12. In this way, comfort of the mask 10 is increased. Further, as the front viewport 20 is transparent, more of the face of the wearer 12 is visible relative to traditional opaque masks, which facilitates non-verbal communication. The facepiece portion 14 is at least partially formed of plastic in this disclosure. The front viewport 20 provides a relatively large, blank surface, that can be customized by adding stickers or decals, depending on preferences of the wearer 12.

(16) The facepiece portion 14 is provided by an inner shell 22 and an outer shell 24, in this example. The inner shell 22 is shown in more detail in FIGS. 3-5. The inner shell 22 includes a forward panel 26 and a perimeter wall 28 projecting from an edge of the forward panel to a free end 30. The forward panel 26 is transparent. In an example, the entire inner shell 22 is made of a transparent material. The perimeter wall 28 is tapered toward the forward panel 26 such that the inner shell 22 can fit within the outer shell 24.

(17) In FIGS. 3-5, the free end 30 is covered by the seal 16. The seal 16 is made of a material configured to form an airtight seal relative to the wearer 12. The seal 16 is made from polyvinyl chloride (PVC) foam in this example, and is adhered to the inner shell 22 using an adhesive.

(18) The perimeter wall 28 includes a first opening 32 in a first side wall 34, a second opening 36 in a second side wall 38 opposite the first side wall 34, and a third opening 40 in a bottom wall 42. A gasket 44 is attached to an exterior of the perimeter wall 28 and surrounds an entire perimeter of each of the first opening 32, second opening 36, and third opening 40. The gasket 44 is made from polyvinyl chloride (PVC) foam in this example, and is adhered to the inner shell 22 using an

adhesive.

(19) The outer shell **24** is shown in more detail in FIGS. **6** and **7**. The outer shell **24** includes a forward panel **46** and a perimeter wall **48** projecting from an edge of the forward panel to a free end **50**. The forward panel **46** is transparent. In an example, the entire outer shell **24** is made of a transparent material. The perimeter wall **48** is tapered toward the forward panel **46** such that the inner shell **22** can fit within the outer shell **24**.

(20) The perimeter wall **48** includes a first opening **52** in a first side wall **54**, a second opening **56** in a second side wall **58** opposite the first side wall **54**, and a third opening **60** in a bottom wall **62**. When the inner shell **22** is inserted into the outer shell **24**, the first openings **32**, **52**, second openings **36**, **56**, and the third openings **40**, **60** are substantially aligned with one another.

(21) The inner and outer shells **22**, **24** are shaped at respective free ends **30**, **50** to correspond to a shape of a face of the wearer **12**. In particular, free ends **30**, **50** are contoured to accommodate a nose of the wearer **12**. Further, the perimeter walls **28**, **48** are dimensioned so that the forward panels **26**, **46** do not directly contact a tip of a nose of the wearer **12** or the mouth of the wearer **12**.

(22) The inner and outer shells **22**, **24** are formed of a plastic material, and in particular a transparent plastic material in this example. The plastic material may be rigid or deformable. The inner and outer shells **22**, **24** are vacuum formed in one example. Other manufacturing techniques come within the scope of this disclosure. The forward panels **26**, **46** could include an anti-fog coating in some examples. In one particular example, an inner surface of the forward panel **26** includes an anti-fog coating.

(23) First openings **32**, **52**, second openings **36**, **56**, and the third openings **40**, **60** are arranged in the perimeter wall **28**, **48** and are configured to permit air to flow in and out of the mask **10**, to and from the mouth and nose of the wearer **12**. A filter **64** (FIGS. **1** and **2**) is arranged between the inner and outer shells **22**, **24** and overlaps the first openings **32**, **52**, second openings **36**, **56**, and the third openings **40**, **60** of the inner and outer shells **22**, **24**. The filter **64** includes a plurality of rear slots **66** and forward slots **68** configured to attach to various structures to hold the filter **64** to the outer shell **24**. The gasket **44** directs air through the filter **64** as opposed to permitting air to leak between the inner shell **22** and the outer shell **24**. The straps **18A**, **18B** keep tension on the outer shell **24** such that the seal **16** is held against the wearer **12**. In turn, the mask **10** is configured such that all air breathed by the wearer **12** passes through the filter **64**. The filter **64** may be made of woven polypropylene fiber and it otherwise of a quality and type such that the mask **10** can use the filter **64** and meet the requirements of an N-95 respirator as defined by the relevant industry standards.

(24) The filter **64** exhibits a length such that a single filter **64** can extend about the mask **10** to overlap each of the first openings **32**, **52**, second openings **36**, **56**, and the third openings **40**, **60**. By providing the filter **64** such that the filter **64** overlaps the first openings **32**, **52**, second openings **36**, **56**, and the third openings **40**, **60** in the perimeter walls **28**, **48**, as opposed to the front viewport **20**, the front viewport **20** remains unobscured by the filter **64**, and the filter **64** is taken out of the direct path of exhalation of the wearer **12**, prolonging the useful life of the filter **64**. The filter **64** protects the wearer **12** from the nearby environmental conditions, and protects those near the wearer **12** from the air expelled by the wearer **12**.

(25) The free end **50** of the outer shell **24** includes a plurality of tabs **70** (FIGS. **6** and **7**) configured to project into corresponding slots **66** (FIG. **2**) in the filter **64** to hold the filter **64** in place, in this example. The mask **10** further includes fasteners **70A-70D** (FIG. **2**) configured to hold strap attachments **72A-72D** relative to the mask **10**. In an example, in addition to holding the strap attachments **72A-72D** relative to the mask **10**, two of the fasteners **70A**, **70D** pass through the forward slots **68** to further hold the filter **64** in place. The fasteners **70A-70D** and strap attachments **72A-72D** may be made of plastic in one example.

(26) FIGS. **8** and **9** illustrate an example fastener **70A** in more detail. It should be understood that the fasteners **70B-70D** are arranged substantially similarly. The fastener **70A** includes a head **74**, a shaft **76**, and a deflectable loop **78**. The shaft **76** exhibits a lesser diameter than the head **74** and the

deflectable loop **78**. Within the deflectable loop **78**, the body of the fastener **70A** includes a cutout **80** such that the deflectable loop **78** can deflect from its relaxed state, which is shown in FIG. 7, to a deflected state in which the deflectable loop **78** is urged toward a centerline of the fastener **70A** under force to permit insertion of the deflectable loop **78** into an opening of the strap attachment **72A**, which permits the opening of the strap attachment **72A** to slide over the shaft **76**. Once the force is removed, the deflectable loop **78** returns to its relaxed state in which the deflectable loop **78** exhibits a diameter greater than the opening of the strap attachment **72A** such that the strap attachment **72A** is retained on the shaft **76**. In use, the head **74** is arranged on an interior of the outer shell **24** such that the shaft **76** projects through an opening **82** in the outer shell **24**. The opening **82** is at a forward corner of the outer shell **24**, which increases stability. The outer shell **24** includes three additional openings at forward corners thereof corresponding to the strap attachments **72B-72D**.

(27) FIG. **10** illustrates an example strap attachment **72A** in more detail. The strap attachments **72B-72D** are arranged substantially similarly. The strap attachment **72A** is configured such that the strap **18A** can be attached to the strap attachment **72A** without tying the strap **18A** into a knot or providing the strap **18A** with a fixed loop, which are rigid loops that do not permit adjustment. In this way, the strap **18A** merely needs to be weaved or wrapped relative to the strap attachment **72A**, which facilitates ease of assembly of the mask **10**. Further, the strap attachment **72A** readily permits a user to make adjustments to the tightness and/or fit of the mask **10** by manipulating the strap **18A**.

(28) The strap attachment **72A** exhibits a length between a forward end **84** and a rear end **86**, and a width perpendicular to the length. The strap attachment **72A** includes a plurality of through-holes, which extend between a first side **88** (FIG. **11**), which here is an underside, and a second side **90**, which here is a top side, of the strap attachment **72A**.

(29) The through-holes that facilitate attachment of the strap **18A** will now be described. The strap attachment **72A** includes a first through-hole **92**, a second through-hole **94**, and a third through-hole **96** configured to permit one to attach the strap **18A** to the strap attachment **72A**. The first through-hole **92** is spaced-apart in a direction of the width of the strap attachment **72A** from the second through-hole **94**. The third through-hole **96** is closer to the forward end **84** than the first and second through-holes **92**, **94**. The third through-hole **96** is also arranged between the first and second through holes **92**, **94**. In this example, the first, second, and third through-holes **92**, **94**, **96** are elliptical and exhibit major axes that are non-parallel to one another. The arrangement of the through-holes facilitates weaving of the strap **18A** such that the strap **18A** can attach to the strap attachment **72A** without tying the strap **18A** into a knot.

(30) In one example, as shown in FIG. **11**, the strap **18A** is attached to the strap attachment **72A** by passing a free end **98** of the strap **18A** from the first side **88** through the first through-hole **92**, and then from second side **90** through the second through hole **94**. Next, the free end **98** is passed from the first side **88** through the third through-hole **96**. The free end **98** is then passed under the segment **100** of the strap **18A** between the first and second through-holes **92**, **94**. Specifically, the free end **98** is passed between the segment **100** and the second side **90**. The free end **98** is then pulled away from the third through-hole **96** to tension the strap **18A**.

(31) The strap **18A** is made of an elastic material in this example such that weaving the strap **18A** in the above-described manner will hold the strap **18A** in place relative to the strap attachment **72A**. An opposite free end of the strap **18A** is weaved in a similar manner relative to strap attachment **72D**, and the strap **18A** can then be passed around the head of the wearer **12**. The strap **18B** is attached to strap attachments **72B**, **72C** in a similar manner. While an exemplary manner for attaching a strap to a strap attachment has been described, this disclosure extends to other strap attachment techniques that do not require a user to tie a strap into a knot.

(32) The strap attachment **72A** includes an indicator arrow **102** in this example. The arrow **102** is pointing in the direction that a user is to pull on the free end **98** to tighten the mask **10**. The arrow **102** may be formed as a through-hole in one example. The arrow **102** may take other forms,

including being provided as a recess into the second side **90**, painted-onto the second side **90**, or applied as a sticker, as examples.

(33) The strap attachment **72A** also includes a through-hole **104** configured to attach to the fastener **70A**. The through-hole **104** is closer to the forward end **84** than the first, second, and third through-holes **92**, **94**, **96**. The through-hole **104** is circular in shape, in this example, and exhibits a diameter that permits the deflectable loop **78** to pass through the through-hole **104** when the deflectable loop **78** is in a deflected state, but not when the deflectable loop **78** is in a relaxed state. The through-hole **104** exhibits a slightly larger diameter than the shaft **76**.

(34) Directional terms such as “upward,” “downward,” “forward,” “rearward,” “top,” “bottom,” etc., are used herein for purposes of explanation and with reference to the orientation of components illustrated in the drawings. Such directional terms should not be considered limiting. Further, it should be understood that terms such as “generally,” “substantially,” and “about” are not intended to be boundaryless terms, and should be interpreted consistent with the way one skilled in the art would interpret those terms.

(35) Although the different examples have the specific components shown in the illustrations, embodiments of this disclosure are not limited to those particular combinations. It is possible to use some of the components or features from one of the examples in combination with features or components from another one of the examples. In addition, the various figures accompanying this disclosure are not necessarily to scale, and some features may be exaggerated or minimized to show certain details of a particular component or arrangement.

(36) One of ordinary skill in this art would understand that the above-described embodiments are exemplary and non-limiting. That is, modifications of this disclosure would come within the scope of the claims. Accordingly, the following claims should be studied to determine their true scope and content.

Claims

1. A mask, comprising: a strap attachment configured such that a strap is attachable to the strap attachment without tying the strap into a knot, wherein the strap attachment includes a through-hole configured to receive a fastener wherein the fastener includes a head, a shaft, and a deflectable loop, and wherein the deflectable loop is biased to a relaxed state in which the deflectable loop is able to retain the strap attachment on the shaft.
2. The mask as recited in claim 1, wherein: the head is arranged between an inner shell and an outer shell of the mask, and the shaft projects through an opening in the outer shell.
3. The mask as recited in claim 2, further comprising: a filter arranged between the inner and outer shells.
4. The mask as recited in claim 3, wherein: the outer shell includes a first opening in a first side wall, a second opening in a second side wall opposite the first side wall, and a third opening in a bottom wall, the inner shell includes a first opening in a first side wall, a second opening in a second side wall opposite the first side wall, and a third opening in a bottom wall such that the first, second and third openings of the inner shell substantially align with corresponding first, second and third openings of the outer shell, and the filter overlaps the first, second, and third openings of the inner and outer shells.
5. The mask as recited in claim 4, wherein the inner and outer shells each comprise: a forward panel; and a perimeter wall projecting from an edge of the forward panel to a free end.
6. The mask as recited in claim 5, wherein the forward panels are transparent.
7. The mask as recited in claim 6, wherein the filter is arranged so as to not obscure a view through the forward panels.
8. The mask as recited in claim 5, wherein: the first, second, and third openings of the outer shell are formed in the perimeter wall of the outer shell, and the first, second, and third openings of the

inner shell are formed in the perimeter wall of the inner shell.

9. The mask as recited in claim 5, further comprising: a seal adjacent the free ends of the perimeter walls of the inner and outer shells.

10. The mask as recited in claim 5, wherein the perimeter walls of the inner and outer shells are tapered toward the forward panels.

11. A mask, comprising: a strap attachment configured such that a strap is attachable to the strap attachment without tying the strap into a knot, wherein the strap attachment is a first strap attachment and the mask further comprises second, third, and fourth strap attachments, wherein the first and second strap attachments are configured to attach to a first strand without tying the first strand into any knots, and wherein the third and fourth strap attachments are configured to attach to a second strand without tying the second strand into any knots.

12. The mask as recited in claim 11, wherein the strap attachment includes a first through-hole, a second through-hole, and a third through-hole configured such that the strap is attachable to the strap attachment by passing the strap through each of the first, second, and third through-holes.

13. The mask as recited in claim 12, wherein: the first through-hole is spaced-apart from the second through-hole in a first direction, and the third through-hole is spaced-apart from the first and second through-holes in a second direction perpendicular to the first direction.

14. The mask as recited in claim 13, wherein the strap attachment is configured such that the strap is attachable to the strap attachment by passing a free end of the strap out of the third through-hole and underneath a segment of the strap between the first and second through-holes.

15. The mask as recited in claim 14, wherein the strap attachment includes an indicator arrow between the first and second through-holes pointing in a direction away from the third through-hole.

16. The mask as recited in claim 12, wherein the first, second, and third through-holes are elliptical and exhibit major axes that are non-parallel to one another.

17. The mask as recited in claim 11, wherein the strap is a strand of elastic material having first and second free ends.

18. The mask as recited in claim 11, wherein the strap attachment includes a through-hole configured to receive a fastener.

19. A mask, comprising: an outer shell including a first opening in a first side wall, a second opening in a second side wall opposite the first side wall, and a third opening in a bottom wall; an inner shell including a first opening in a first side wall, a second opening in a second side wall opposite the first side wall, and a third opening in a bottom wall such that the first, second and third openings of the inner shell substantially align with corresponding first, second and third openings of the outer shell; and a filter overlapping the first, second, and third openings of the inner and outer shells, wherein the inner and outer shells each comprise a transparent forward panel and the filter is arranged so as to not obscure a view through the forward panels.
